USN | | | | | | | | | | | [23TCSE402]

FOURTH SEMESTER B.Tech EXAMINATION, FEBRUARY - 2025

DESIGN AND ANALYSIS OF ALGORITHMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART - A			**5 x2=10**
1.	**a.**	What are the two phases of performance evaluation?	L1	CO1	**02**
	b.	Differentiate divide and conquer and greedy method.	L4	CO2	**02**
	c.	Define dynamic programming.	L2	CO3	**02**
	d.	Describe the role of c(-) and u(-) functions in branch and bound to find solutions tuples.	L3	CO4	**02**
	e.	What does NP stand for, and what is the significance of this class in computational theory?	L2	CO5	**02**
		PART – B			**6 X 5=30**
2.	**a.**	How do describe the algorithms in natural languages? Explain the structure of pseudo code for various algorithm.	L3	CO1	**05**
		OR			

	b.	Estimate the Time complexity for the following code
		int a[n];
		for(int i = 0;i < n;i++)
		cin = a[i]
		for(int i = 0;i < n;i++)
		for(int j = 0;j < n;j++)
		if(i!=j && a[i]+a[j] == z)
		return true
		return false
			L5	CO1	**05**

3.	**a.**	Discuss Linear search with an example and analyze the Best, Average and Worst case for the same input of your example.	L4	CO2	**05**
		OR			
	b.	Write an algorithm of Merge sort for the given list L={2,12,18,3,34,27,56,4,8,3,10} using Divide and conquer method and analyze the performance of the given list.	L3	CO2	**05**
4.	**a.**	Discuss about the algorithm and pseudocode to find the minimum spanning tree using prims algorithm. Find the minimum spanning tree for the graph given below and discuss the efficiency of algorithm.	L5	CO3	**05**

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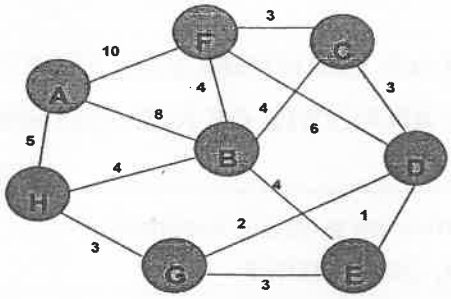
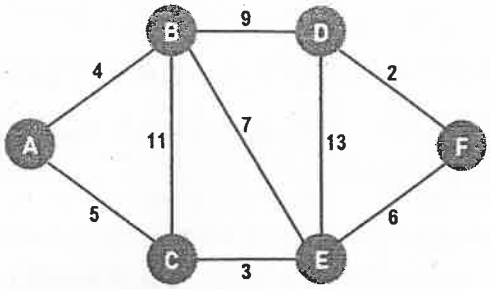
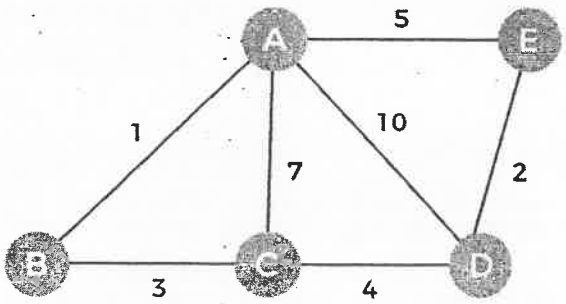
					
		OR			
	b.	Demonstrate how Backtracking technique can be used to solve n-Queens problem (n = 4).	L3	CO3	**05**
5.	**a.**	Write an algorithm scheme LC Branch and bound which searches for a least-cost answer node and explain.	L2	CO4	**05**
		OR			
	b.	Examine Travelling salesperson using Branch and Bound.	L3	CO4	**05**
6.	**a.**	How is class P related to class NP? What are the key differences between these two classes, and how do they impact the design of algorithms?	L4	CO5	**05**
		OR			
	b.	What is the formal statement of Cook's Theorem? Explain it in simple terms, particularly in relation to decision problems and computational complexity?	L2	CO5	**05**
7.	**a.**	Sort an element using insertion sort for the list of elements {23,1,10,5,2} and write pseudocode for insertion sort.	L3	CO2	**05**
		OR			
	b.	Discuss the Subset-Sum Problem and outline how the Backtracking method can be used to solve it. Provide a simple example to illustrate the process.	L4	CO3	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Define performance analysis. Compute the performance for iterative sum and recursive sum through step count method.	L2	CO1	**10**
		OR			
	b.	$f(n)=O(g(n))$, $f(n)=\Omega(g(n))$ and $f(n)=\Theta(g(n))$, illustrate these relations in estimating the time complexities with suitable examples.	L3	CO1	**10**
9.	**a.**	Describe the importance of pivot element in quick sort. How does the position affect the performance of quick sort? Explain with algorithm.	L2	CO2	**10**
		OR			
	b.	Infer Breadth First Search and Traversal with a suitable example	L2	CO2	**10**
10.	**a.**	Compute the shortest path algorithm using Dijkstra's algorithm starting at the node A. Write pseudo code for the same.	L3	CO3	**10**

					
		OR			
	b.	Explain the Traveling Salesperson Problem, and how is it solved using Dynamic Programming?	L2	CO3	**10**
11.	**a.**	Demonstrate 0/1 knapsack problem using Least Cost Branch and Bound.	L3	CO4	**10**
		OR			
	b.	Illustrate FIFO Branch and Bound with example.	L3	CO4	**10**
12.	**a.**	Briefly explain the classes NP-hard and NP-complete?	L2	CO5	**10**
		OR			
	b.	What is parameterized complexity in the context of the Clique Decision Problem, and how can it be applied to improve the efficiency of solving this problem for specific values of k?	L4	CO5	**10**
13.	**a.**	Apply divide and conquer to search an element 'k' in an array a[1:n] using binary search and explain the algorithm.	L3	CO2	**10**
		OR			
	b.	Calculate the Minimum cost of the following weighted graph using Kruskal's algorithm. 	L5	CO3	**10**